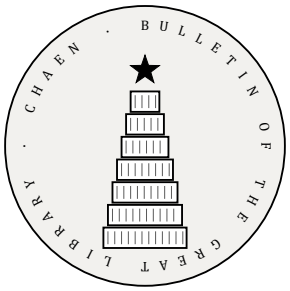
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THE DECIPHERMENT OF THE THIRD SCRIPT OF THE BABEL RELIC

By APPLEBY

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By Appleby*

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Summary. In the scrolls of the Babel Relic, beside the two scripts that can be read, there is a "third script" which no one since Severin has been able to read. It is shown here to be an archaic form of the thirty-six thousand characters: a sign is for the most part an old form of a single character with its strokes compressed, and in a few cases a joining of two or three characters; and its rules of combination, when the present order is reversed, agree with Roxburgh's grammar of the characters. The inference engine went through every surviving fragment of the third script in the Relic: from a thousand of them it proposed the correspondences of form, which were checked one by one by the author and scholars of the Tower of Tongues; and when the table of correspondences had been fixed, the other four hundred and twelve were read by it. The checking had three standards: the correspondences must account for the phrases and rules of combination that recur in the fragments; what is read must agree with the known parallel passages; and they must read through the fragments that took no part in drawing up the table of correspondences. All three are met: of 1,148 recurring phrases, 1,102 read as lawful combinations of clear sense; all 27 parallel passages agree; and of 412 fragments that took no part in drawing up the table, 388 are read through. The dangerous fragments were handled only in safe transcriptions, in an authorised isolated room, and were not added to the materials of the general engine. Such contents as were read and belong to the Starlight grade or above have been reported to the Great Library for grading, and are not printed here.

1. Introduction

Severin's Twenty-Fourth Question asks: "Can the third script in the scrolls of the Babel Relic be read?" [1] Two scripts in the scrolls of the Relic have always been read: the thirty-six thousand characters in their present form, and the script in which the people of Babel wrote their records. The third script lies among them, sometimes filling a whole scroll, sometimes only a few lines. The copyists copy it stroke by stroke as they copy the rest, but do not know what they copy.

Three roads have been tried. One is to compare it with the characters: its signs are like none of the thirty-six thousand, and the comparing has yielded only some tens of seeming likenesses. Another is to compare it with other ancient scripts: after Llewellyn had set out the descent of the scripts [3], some placed it in a branch of the primal runes, but read according to that branch it yields no sentence. The third is to count it: its signs are more than two thousand kinds, more than any script that writes sounds and far fewer than the thirty-six thousand characters, so that some have held it to be a lost tongue, of which the twenty-odd parallel passages noted in the catalogues of the Relic are only translations, too few to read the rest by.

This paper takes the first road, but compares in another way: not the single forms, but the places where they occur. Lynn's correspondence formula can find, in a very

*The Great Library, Chaen; formerly of Silverwing College.

long text, the places that are related to one another [6], and the inference engine that has appeared this year can verify across volumes [7]. I asked the engine to compare, for every sign of the third script, the places where it occurs, the signs before and after it, and the signs with which it occurs together, with the same distributions of the thirty-six thousand characters in known texts, and to propose candidates of the form "this sign is that character". The candidates are not settled by the engine; they are checked one by one by men.

The conclusion of this paper is that the third script is an archaic form of the thirty-six thousand characters.

2. Materials

There are in the Relic 1,412 surviving fragments containing the third script; see Table I. Of these, 1,187 have been brought down by assessment of safety to the Dusk grade or below; 225 remain in the Starlight grade, and by the charter were read only in the Starlight Vault: those who handled them all hold the higher academic standing of the Silver Society and the written leave of the head of the Library; what was used was the safe transcriptions made by the Tower of Tongues within the Vault, in which signs that carry mana are kept only as codes, and which do not leave it; and the engine that read them was a separate one set up in the Vault, not joined to the general engine, whose readings and records are kept at the Starlight grade and are not added to the materials of the general engine. No fragment of the Abyss grade was used.

TABLE I. The materials

Kind	*Number*	*Use*
Fragments of the Dusk grade or below	1,187	1,000 for drawing up the table, 187 kept for the trial
Safe transcriptions of Starlight fragments	225	all kept for the trial, read only in the Starlight Vault
Total	1,412	
Signs of the third script	2,904 kinds	318,600 occurrences

The thousand fragments used for the table were put into codes by the Great Library's Code Table [5]; signs that carry mana are, as always, not copied in their own forms, but kept only as codes. The 412 fragments kept for the trial had been read neither by the engine nor by those who checked, before the table was drawn up; nor were they, or any copies of them, among the archives of the Code Table on which the first engine was trained, for the third script of the Relic has been kept only in the scrolls themselves and in the copies of the Tower of Tongues, and was never taken into those archives; and the four scholars who checked with me had not handled them before. Sixty-one of them were released, being brought down from the Starlight grade to the Dusk, only after the table had been fixed; they are counted among the 187 kept for the trial, Table I giving the grades as they stood at the trial. Release by bringing down the grade is, by the charter, proposed by the keeper of the hall and decided by the council of the Library; the fragment in which Stafford read the proof of the Fourth Question was released in the same way [8].

3. What the engine did

In drawing up the table the engine did one thing only: it compared distributions and proposed candidates. For each sign it reckoned the places where the sign occurs in the fragments, the signs that commonly come before and after it, and the number of times it occurs together with other signs; it then sought in the known texts of the thirty-six thousand characters the characters whose distributions are nearest, ranked several candidates by their nearness, and attached to each the passages of the fragments and the comparisons on which it was proposed [7]. By the distribution is meant the three signs on either side, counted without regard to the side on which they stand, the place in the line, and the number of times a sign stands in one line with others; the known texts are the old copies, from before the First Era, gathered in the Code Table, and not the scrolls of the Relic. The joinings of two or three characters were proposed in the same way, the engine taking two or three characters that commonly stand together in the present writing as one, and comparing their distribution. This year's report proposed that whatever an inference engine gives should carry sources and derivations that can be checked; this paper has followed that proposal, and all the candidates with their sources are kept in the Tower of Tongues, where they may be consulted according to the grades of the fragments they touch.

The candidates were checked by men. Four scholars of the Tower of Tongues and I checked them one by one: a candidate was written into the table only if it accounted for the recurring phrases and rules of combination in which it stood, which is the first standard of Section 5. The engine proposed 3,871 candidates in all, of which 2,611 were written into the table; the 1,260 not written in were for the most part second choices for the same signs. Among those it proposed were many that looked well but failed when checked: some only because the two signs both stood commonly at the head of a sentence; some whose sources could not be found, or were found but did not say what was claimed. When the table had been fixed, it was set as a whole against the parallel passages and read upon the fragments kept for the trial, which are the second and third standards. Both these materials were held back while the table was drawn up: the copies of the parallel passages in the present characters are in the Relic, and so were never among the known texts with which the engine compared, nor did those who checked turn to them; and the fragments kept for the trial, as was said above, had been read by no one. Every correspondence in the table has passed the first; those that occur in the parallel passages or in the fragments kept for the trial had to pass the second and the third as well, and any found against them there would have been struck out; not one was.

4. Forms and combinations

The checked table shows several features of the script.

First, a sign is for the most part an old form of one character. The old forms have fewer strokes than the present ones, as though several strokes of the present form had been drawn into one and the places of the parts had been moved. Each of the 2,611 checked signs has one checked reading only: 2,437 correspond each to one character, and 174 are joinings of two or three characters, like the joined writing of today. Among

the 2,437 there are ninety-eight pairs that are two old forms of one character, so that they answer to 2,339 different characters.

Secondly, its rules of combination are the reverse of today's. Roxburgh's grammar of the characters [2] lays down the lawful combinations of today: the qualifying character first, the qualified after. The third script writes them the other way about. What is reversed is only the order of the qualifying and the qualified within a combination; the order of the combinations in a line, and the order in which the four elements are named, are as today. When its combinations are reversed, they accord with Roxburgh's grammar. It was for this reason that those before, comparing in today's order, could find no likeness.

Thirdly, its signs are fewer than today's characters. One reason is that these fragments write of only so many things; another, it may be, is that some characters did not yet exist. Of the 2,339 characters so checked, fewer than half are in common use today; many are rare today, or have been seen only in old inscriptions.

Let one sign show how a correspondence was settled. The sign coded T-0417 in the Code Table occurs 1,926 times in the fragments used for the table, sixty-two times in a hundred at the head of a line, and is commonly followed by several signs of another family; it stands in the same line with the three signs afterwards read as Oss, Vo and Flame 112 times. Among the candidates the engine ranked for it, the first was Eaam, and the second another character that commonly opens a calling. By the first standard, the forty-one recurring phrases in which it stands all become lawful invocations when it is read as Eaam, while read as the second, seventeen of them break the grammar; and T-0417 was written into the table as Eaam. The three signs read as Oss, Vo and Flame were each settled by the phrases in which they themselves stand, and not by T-0417. When the table had been fixed, it was found to stand, in the nine parallel passages of invocations, exactly where the present characters write Eaam; and among the fragments kept for the trial there are seventeen with lines calling upon the four elements, in all of which it reads through. The reversed order of combination was found in the same way: of the pairs of candidates the engine proposed, eight or nine in ten ran contrary to the order of today's combinations, and only when they were reversed did the checked combinations all agree with Roxburgh's grammar.

Fig. 1 groups the signs into twelve families by the parts they share, and the characters into ten classes by their parts, both groupings being fixed before the table was drawn up, and the classes set in order, after the table was fixed, by the family to which each most corresponds; the shade of each cell is the number of checked correspondences between the two, a joining being counted by the first character of its reading. The signs of each family correspond for the most part to one or two classes of characters, which agrees with their coming from one set of parts.

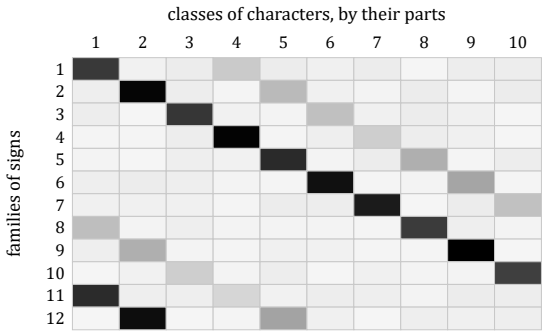


Fig. 1. The 2,611 checked correspondences, by family of signs (rows) and class of characters (columns); the darker the cell, the more correspondences

5. The three checks

Table II gives the results of the three standards.

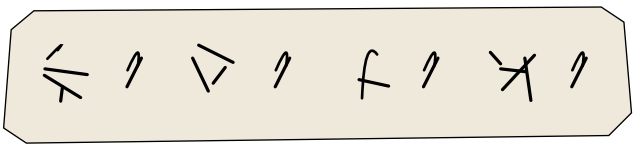
Table II. The results of the three checks

Standard	*What was checked*	*Result*
1. To account for recurring phrases and rules of combination	1,148 phrases occurring three times or more	1,102 read, lawful and of clear sense (96 in a hundred)
2. To agree with known parallel passages	27 parallel passages	all agree
3. To read through the fragments that took no part in the table	412 fragments kept for the trial	388 read through, 19 more than half read, 5 not read

The first standard. Of the 1,148 recurring phrases, 1,102 when read by the table accord with Roxburgh's grammar and their sense is clear. Of the other forty-six, thirty-one contain signs not yet read; fifteen can be read and are lawful, but their sense is obscure. Not one reads against the grammar.

The second standard. A parallel passage is one of which the Relic holds another copy written in the present characters; which answers to which was noted long ago in the catalogues of the Relic, by the titles in the present characters at the heads of the scrolls, and was not picked out after the reading, and the catalogues note these twenty-seven and no others. Of the twenty-seven, nine are invocations of activation, such as the four lines calling upon earth, wind, water and flame, where the signs of the third script answer exactly to Eaam, Oss, Vo and Flame [2]; seven are the names and order of the twelve mana moons; six are pieces of the Hymn of Creation; and the other five are catalogues and colophons. Fig. 2 shows the line calling upon the four elements on one fragment, with its reading. All twenty-seven agree, and not one required the table to be altered in order to agree. The nine invocations and the seven namings of the moons, however, are formulas known to all, and phrases of the same kind were among those of the first standard, so that their agreement weighs the less; the six pieces of the Hymn and the five catalogues and colophons are longer and are no formulas, and their agreement weighs the more.

The third standard is the most important: had the table merely been fitted together, it would hardly read the fragments it had never seen. The manner of dividing the lines, of opening the joinings and of reversing the order was fixed before the trial and not changed during it; and whether a fragment reads or not was judged by two scholars of the Tower of Tongues who had taken no part in drawing up the table. Of the 412 fragments kept for the trial, 388 are read through, that is, more than nine in ten of the signs that remain can be read and the sense runs on; 19 are read more than half, but not through; and the five not read are all fragments so damaged that only some ten or twenty signs remain. Of the sixty-one released only after the table had been fixed, fifty-nine are read through, and the other two more than half. For comparison a table falsely matched was also tried: the checked correspondences were exchanged at random among signs of one family that occur about equally often, and the 412 fragments read by it; not one could be read through.



Eaam · *Oss* · *Vo* · *Flame* ·

signs read by the table; the dots are the dividers of the line

FIG. 2. The line calling upon the four elements on one fragment (above, the signs of the third script) and the present characters read from it by the table (below)

6. What was read, and what is not printed

The fragments read through are for the most part descriptions of laws, orders of rites, catalogues and colophons. Part of their contents belongs to the Starlight grade or above: by section 18 of the Law on Research and the Keeping of Privileged Knowledge it was reported to the Great Library for grading within ninety days [10], and by section 21 of the League Heritage Law this paper was sent, before printing, to the Great Library and the Commission for the Control of Forbidden Arts for review [11]. The examples and the figures given in this paper are all taken from the parts the review allowed to be printed: the only signs it allowed to be shown are those of the line of Fig. 2, and only a few codes are named in the text; the whole table of correspondences is kept in the Tower of Tongues according to the grades of the fragments it touches. The counts the review allowed to be printed as they are. The whole of what was read is kept in the Tower of Tongues, to be consulted according to the grade fixed.

7. What remains

Two hundred and ninety-three signs have not yet been read; together they make only eight in a thousand of all occurrences, and most occur only a few times.

Some signs of the third script carry mana. Llewellyn says that within one branch of writing, the later the form, the weaker the mana it carries [3]; on the reading of this paper the third script is an older branch of the thirty-six thousand characters, and that it carries mana is no wonder. But whether the thirty-six thousand characters descend from it, or both come from a still older branch, this paper cannot decide.

Whether the characters suffice, that is, the first half of Severin's Tenth Question, of which Monkton settled only a part [4], this paper does not touch: nothing in what has been read of the third script is beyond the present characters to write; but this shows only that what has been read can be written, says nothing of the 293 signs not read, and does not show that every law can be. The second half, whether there are superfluous characters, Merrick has proved [9]; in the third script the ten or so superfluous characters each have their old forms as well, so that they were already in use in old times.

Acknowledgments. Four scholars of the Tower of Tongues checked every candidate with me; the engine rooms of the Kanaich Institute of Technology and of the Great Library lent the spare capacity of the inference engine; and the keepers of the Babel Relic made the safe transcriptions of the Starlight fragments, and kept the order of the fragments held back for the trial.

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